

School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Lecture-23 Scribe Submission

Case Study: Amazon Black Friday Sale 2026 (Coding Session – Tails and Bounds)

Course: CSE400

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1 Lecture Theme / Core Focus

Lecture 23 introduces the idea of reasoning about uncertainty in computing systems through a case study based on Amazon Black Friday Sale 2026. The lecture connects a large-scale system event (high demand during a sale) with tail bounds and system reliability, indicating that the purpose is to understand how probability inequalities can help estimate or bound rare or extreme events in computing systems.

The title itself clearly frames the lecture as:

- Predicting the impossible
- Using tail bounds
- Studying system reliability
- Through a case study
- In a coding session focused on tails and bounds

2 Context of the Case Study: Amazon Black Friday Sale 2026

The PPT explicitly states that the case study is built around the Amazon Black Friday Sale 2026. It notes:

- The sale is expected to officially begin on Friday, November 27, 2026
- Official confirmation from Amazon is typically closer to the date
- The date is inferred using historical patterns and expert forecasts
- The lecture references forecast sources such as:
 - 91mobiles
 - Bajaj Finserv

Interpretation from the Slide Text (Within PPT Boundaries)

The lecture uses this event as a real-world example of a situation where:

- Demand can spike unpredictably
- System load can become extreme
- Computing infrastructure must handle uncertainty
- Rare high-load events matter for reliability analysis

This interpretation follows directly from the lecture title and the slide phrase “*Connect with Uncertainty and Computing*”, which links the Amazon sale scenario to probabilistic system analysis.

3 Central Conceptual Bridge: “Connect with Uncertainty and Computing”

A slide explicitly contains the phrase:

Amazon Black Friday Sale 2026 – Connect with Uncertainty and Computing

This is the conceptual bridge of the lecture.

Meaning of this Bridge in the Lecture Flow

The lecture appears to move from:

- A real-world high-stakes event (Amazon sale)
- To the need to model uncertainty
- To mathematical tools that help reason about extremes
- To implications for system reliability

So the lecture is not merely about a sale event; it uses the event as a computing systems reliability scenario under uncertain demand.

4 Main Mathematical / Analytical Tools Mentioned

The extractable PPT text explicitly lists three tail-bound tools:

- Markov’s Inequality
- Chebyshev’s Inequality
- Chernoff Bound

These are the only explicitly named theoretical results visible in the PDF text, so they form the core theoretical structure of the lecture.

5 Exam-Ready Reconstruction of Lecture Flow

5.1 Motivation

The lecture begins with a high-impact, realistic computing event:

- Amazon Black Friday Sale 2026
- Large-scale online demand
- High uncertainty in traffic / load
- Need for robust systems under rare extreme events

Why this Matters

In system reliability, average behavior is often not enough.

What matters is the chance of extreme outcomes (for example, unusually high demand or stress). This is exactly why the lecture turns to tail bounds. This reasoning is directly supported by the lecture title and the listed inequalities/bounds.

5.2 Transition to Tail Bounds

The phrase “*Predicting the Impossible*” strongly suggests that the lecture focuses on estimating or bounding events that are:

- Rare
- Extreme
- Hard to predict exactly
- But still important for system design and reliability

Instead of exact probabilities from full distributions, the lecture likely emphasizes bounds on unlikely events.

This is consistent with the listed tools:

- Markov’s Inequality → coarse upper bound
- Chebyshev’s Inequality → variance-aware bound
- Chernoff Bound → stronger concentration bound for sums / deviations

Note: I am intentionally not writing formulas, because they are not visible in the extractable PPT text you provided, and you explicitly said not to add anything outside the PPT.

6 Topic-Wise Scribe

6.1 Markov’s Inequality

What is Explicitly Present in the PPT

The term “*Markov’s Inequality*” is explicitly listed.

Lecture-Role Reconstruction (Strictly Bounded)

From its placement among the listed tail bounds, Markov's Inequality is presented as one of the first tools for bounding rare events in uncertain systems.

Exam-Ready Takeaway

- Markov's Inequality is introduced as a tail-bound method
- It is relevant when trying to reason about extreme outcomes
- In the context of the Amazon sale case study, it supports the theme of system reliability under uncertainty

What Cannot Be Safely Added

Because the visible PDF text does not show the formula, assumptions, proof, or worked example, those cannot be included without violating your instruction.

6.2 Chebyshev's Inequality**What is Explicitly Present in the PPT**

The term "*Chebyshev's Inequality*" is explicitly listed.

Lecture-Role Reconstruction (Strictly Bounded)

Chebyshev's Inequality appears as the next tool in the progression of tail bounds, suggesting a refinement or another way to bound deviations in uncertain settings.

Exam-Ready Takeaway

- Chebyshev's Inequality is part of the lecture's tail-bounds toolkit
- It is used in the broader discussion of uncertainty in computing
- It contributes to reasoning about system reliability in the Amazon Black Friday scenario

What Cannot Be Safely Added

No explicit theorem statement, notation, proof, or example is visible in the extracted PDF text, so none should be inserted.

6.3 Chernoff Bound**What is Explicitly Present in the PPT**

The term "*Chernoff Bound*" is explicitly listed.

Lecture-Role Reconstruction (Strictly Bounded)

Chernoff Bound is included as the third named analytical tool, implying the lecture likely builds toward a stronger or more refined concentration-style bound for system reliability analysis.

Exam-Ready Takeaway

- Chernoff Bound is a major tail bound highlighted in this lecture
- It is part of the mathematical reasoning used to understand extreme system events
- It is explicitly tied to the lecture's theme of "*Predicting the Impossible*" in the context of Amazon Black Friday Sale 2026

What Cannot Be Safely Added

Since the extractable text does not reveal formulas, conditions, or derivation steps, they must not be added.

7 Logical Structure of the Lecture (Clear Flow)

Based strictly on the visible slide text, the lecture's flow can be reconstructed as:

1. Introduce a real-world uncertain event

- Amazon Black Friday Sale 2026
- Expected sale date: Friday, November 27, 2026

2. Connect event to computing

- "Connect with Uncertainty and Computing"

3. Motivate system reliability

- High demand / uncertain behavior implies reliability concerns
- Extreme events matter more than average events
- Hence need for probabilistic tail reasoning

(This is supported by the lecture title and slide wording.)

4. Introduce tail-bound tools

- Markov's Inequality
- Chebyshev's Inequality
- Chernoff Bound

5. Conduct coding / class participation session

The lecture is explicitly labeled:

- **Coding Session – Tails and Bounds**
- **L23 : Class Participation** (appears twice in the extract)

This strongly indicates that the class likely included:

- Interactive reasoning
- Participation-based problem exploration
- Possibly implementation or computational illustration of tail bounds

However: since no code or examples are visible in the extractable text, no code content should be fabricated.

8 Definitions / Notation / Assumptions / Theorems / Proofs / Examples (Strict Compliance Section)

Since you specifically asked for these if available in the PPT, here is the exact compliance summary:

8.1 Definitions and Notation

Not explicitly visible in the extractable PDF text.

Therefore: cannot be included safely.

8.2 Assumptions and Conditions

Not explicitly visible in the extractable PDF text.

Therefore: cannot be included safely.

8.3 Statements of Theorems / Results

Only the names of the results are visible:

- Markov's Inequality
- Chebyshev's Inequality
- Chernoff Bound

Full theorem statements are not visible in the extractable text.

8.4 Proofs / Proof Sketches

Not visible in the extractable PDF text.

Therefore: cannot be included.

8.5 Worked Examples with Intermediate Steps

Not visible in the extractable PDF text.

Therefore: cannot be included.

9 Class Participation / Coding Session Note

The lecture explicitly includes:

- (Coding Session – Tails and Bounds)
- L23 : Class Participation

Exam-Ready Interpretation

This indicates the lecture was not purely theoretical. It likely involved:

- Active student engagement
- Some form of coding-based illustration
- Hands-on application of tail-bound ideas

But because the uploaded PDF's extractable content does not show the actual code or tasks, the scribe should only state that a coding session and class participation component were part of the lecture, without inventing details.

10 Final Exam-Ready Summary (Short Version)

Lecture 23 Summary

Lecture 23, titled “*Predicting the Impossible: Tail Bounds and System Reliability – Case Study: Amazon Black Friday Sale 2026*”, uses the upcoming Amazon Black Friday Sale 2026 as a real-world case study to connect uncertainty in computing systems with system reliability. The lecture frames the sale as an example of a large-scale uncertain event where extreme demand can affect system behavior. To reason about such rare or extreme outcomes, the lecture highlights three key probabilistic tools: Markov’s Inequality, Chebyshev’s Inequality, and Chernoff Bound. The session is explicitly marked as a coding session on tails and bounds and also includes class participation, indicating an interactive or applied treatment of these concepts. Since the extractable PPT text does not reveal formulas, proofs, or worked examples, the exam-ready reconstruction should remain limited to the lecture’s visible conceptual structure and named results.